LUCIDMSK CME PROGRAM

MODULE 07

Ankle & Foot MRI: Ligaments, Tendons & Fracture Classification

ATFL/CFL/PTFL Grading · Sanders CT · Weber/Lauge-Hansen · Achilles · Charcot · Turf Toe · Freiberg

1.0 AMA PRA Cat 1 | **Ankle** | **CME Article Review**

LEARNING OBJECTIVES

1. Grade lateral ankle ligament injuries (ATFL, CFL, PTFL) on fluid-sensitive MRI.
2. Apply the Sanders CT classification to intra-articular calcaneal fractures (Types I–IV).
3. Apply the Weber and Lauge-Hansen classifications to ankle fractures and correlate with syndesmotic injury.
4. Recognize Achilles tendinopathy, partial tears, and complete tears on sagittal MRI, and identify spring ligament/posterior tibialis tendon dysfunction.
5. Apply Eichenholtz staging to Charcot neuroarthropathy (Stages 0–III).
6. Classify turf toe and Freiberg infraction using established grading systems.

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1. Lateral Ankle Ligament Injury Grading

The lateral ankle ligament complex — anterior talofibular ligament (ATFL), calcaneofibular ligament (CFL), and posterior talofibular ligament (PTFL) — is injured in a predictable sequence during inversion injury, with the ATFL failing first, followed by the CFL with increasing force, and the PTFL only in the most severe injuries. Grading each ligament independently on fluid-sensitive sequences directly informs whether conservative management or surgical stabilization is indicated.

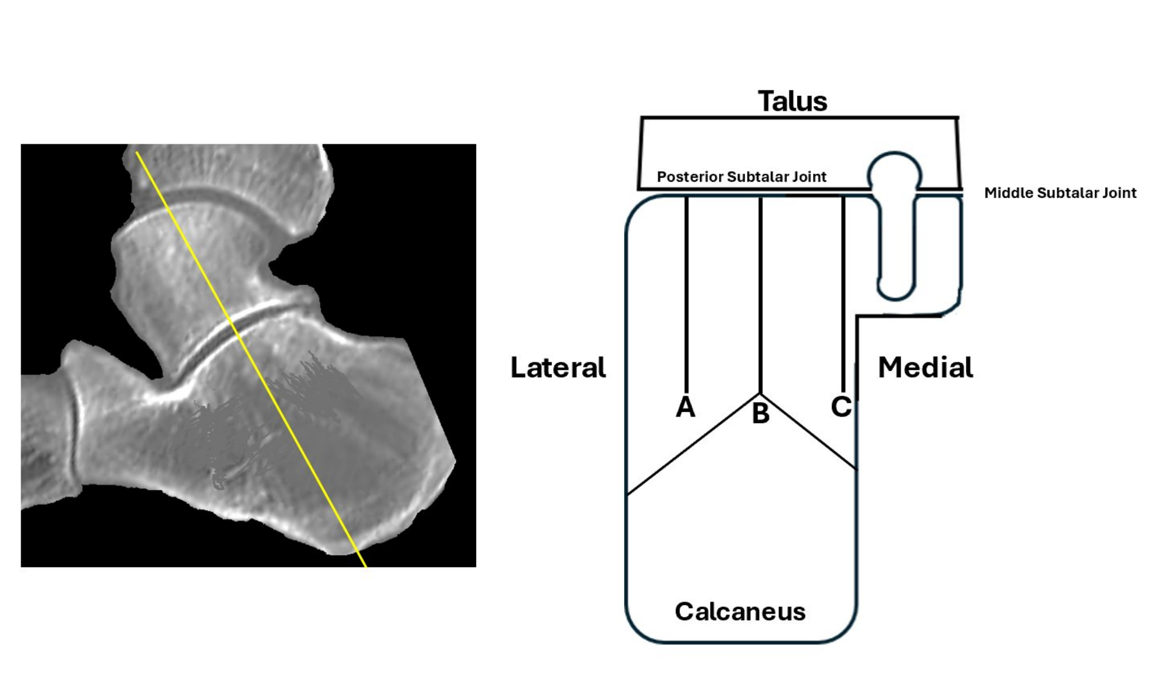
Lateral Ankle Ligament Injury Grading

Grade	Finding	Management
I (Sprain)	Ligament thickened/edematous; fibers continuous; no laxity	Conservative; functional rehabilitation
II (Partial tear)	Partial fiber discontinuity; ligament attenuated but grossly intact	Conservative; brace/boot, longer recovery
III (Complete tear)	Full-thickness discontinuity; ligament non-visualized or frankly disrupted	Surgical assessment; may require reconstruction if chronic instability develops

- Always assess the ATFL, CFL, and PTFL independently rather than reporting "lateral ligament complex injury" as a single entity — the specific combination of ligaments involved, not just overall severity, determines whether the ankle is mechanically unstable.

2. Calcaneal Fractures — Sanders Classification

Intra-articular calcaneal fractures are classified by the Sanders system, based on the number and location of fracture lines through the posterior facet on coronal CT — the imaging plane in which this fracture pattern is best appreciated. The classification divides the posterior facet into three columns (A, lateral; B, central; C, medial), and the number of fracture lines through these columns directly predicts surgical complexity and prognosis.



Sanders classification reference: the posterior facet is divided into lateral (A), central (B), and medial (C) columns on coronal CT.

Sanders Classification — Calcaneal Fractures

Type	Description	Management
I	Non-displaced, regardless of the number of fracture lines	Conservative treatment
II (A/B/C)	Two-part fracture of the posterior facet — a single fracture line through column A, B, or C	ORIF indicated
III (AB/AC/BC)	Three-part fracture — two fracture lines, with a centrally depressed fragment	ORIF; worse prognosis than Type II
IV	Four-part, comminuted fracture of the posterior facet	Primary subtalar fusion often considered

Key References

• Sanders R, Fortin P, DiPasquale T, Walling A. Operative treatment of displaced intraarticular calcaneal fractures. J Bone Joint Surg Am 1992.
• Rammelt S, Zwipp H. Calcaneus fractures: facts, controversies and recent developments. Injury 2004.

3. Ankle Fracture Classification & Syndesmotic Injury

Two complementary systems are used for ankle fractures. The Weber (Danis-Weber) classification is purely anatomic, classifying fractures by the level of the fibular fracture relative to the syndesmosis — simple and highly reproducible. The Lauge-Hansen system is mechanism-based and predicts the sequence of ligamentous and osseous injury, which is particularly useful for correlating MRI findings of ligament or syndesmotic disruption with the radiographic fracture pattern.

Weber (Danis-Weber) Classification

Type	Fibular Fracture Level	Stability
A	Below the level of the syndesmosis	Stable; syndesmosis intact
B	At the level of the syndesmosis	Variable — depends on subtype and medial column integrity
C	Above the level of the syndesmosis (including Maisonneuve fracture)	Unstable; implies syndesmotic disruption

Distal fibular fracture with ankle mortise widening on radiograph — the imaging pattern that should prompt explicit assessment of syndesmotic integrity.

Syndesmotic widening is assessed by measuring the tibiofibular clear space on axial images approximately 1cm above the tibial plafond. A clear space greater than 6mm, or side-to-side asymmetry greater than 2mm, indicates diastasis and syndesmotic disruption, and should prompt careful evaluation of the anterior and posterior inferior tibiofibular ligaments.

Syndesmotic Width — Reference Values

Measurement	Value
Normal tibiofibular clear space	< 6mm (axial MRI, 1cm above plafond)
Diastasis	> 6mm, or side-to-side asymmetry > 2mm

Maisonneuve fracture: proximal fibular fracture, often with a deceptively normal-appearing distal ankle on standard radiographs.

! A Maisonneuve fracture — a proximal fibular fracture with an intact-appearing distal ankle on standard ankle radiographs — is a classic Weber C-equivalent injury pattern that requires imaging of the full length of the fibula and explicit assessment of the syndesmosis; it is easily missed if the study is limited to the ankle alone.

Key References

- Lauge-Hansen N. Fractures of the ankle II: combined experimental-surgical and experimental-roentgenologic investigations. Arch Surg 1950.
- Hermans JJ, Beumer A, de Jong TA, Kleinrensink GJ. Syndesmotic ankle injuries: MRI measurement and classification. Eur J Radiol 2012.

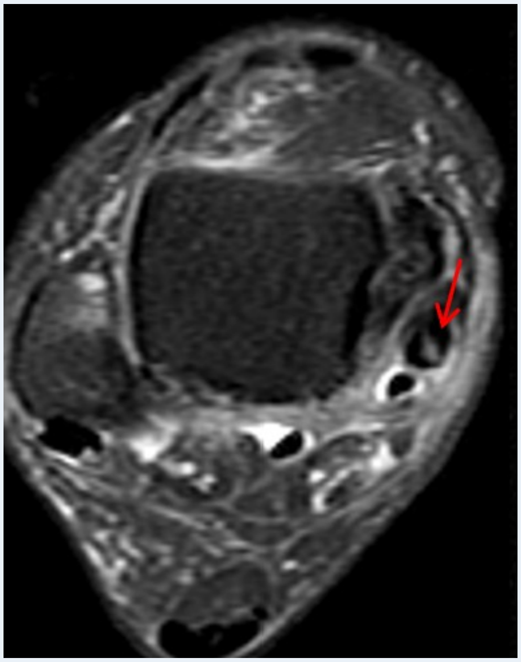
4. Achilles Tendinopathy, Spring Ligament & Sinus Tarsi Syndrome

Achilles pathology spans a spectrum from tendinosis to partial and complete tears, best assessed on sagittal fluid-sensitive and T1 sequences, with attention to Kager's fat pad for secondary signs of tendon retraction. Separately, the spring ligament (superomedial calcaneonavicular ligament) acts as a sling supporting the talar head, and its failure — strongly associated with posterior tibialis tendon dysfunction — is a key driver of adult acquired flatfoot deformity.

Achilles Tendon Pathology Spectrum

Finding	MRI Appearance
Tendinosis	Fusiform thickening; intrasubstance increased T2 signal without discrete fiber discontinuity
Partial tear	Focal fiber discontinuity involving a portion of the tendon cross-section; tendon remains in continuity
Complete tear	Full-thickness discontinuity; retracted tendon ends; Kager's fat pad fills the resultant gap

Sinus tarsi syndrome presents on MRI as replacement of the normal fat signal within the sinus tarsi by low T1/high T2 signal (fibrosis or ganglion-type tissue), accompanied by loss of the cervical and interosseous talocalcaneal ligaments on fluid-sensitive sequences. Clinically, this follows an inversion injury and presents as lateral hindfoot pain.



Axial MRI demonstrating posterior tibialis tendinopathy/tearing — a finding that should always prompt deliberate evaluation of the spring ligament for associated injury.

- A spring ligament tear should always prompt deliberate evaluation of the posterior tibialis tendon — the spring ligament functions as the sling supporting the talar head, and its failure together with PTT dysfunction is the combination that drives progressive collapse of the medial longitudinal arch in adult acquired flatfoot deformity.

Key References

- Theobald P, Bydder G, Dent C, et al. The functional anatomy of Kager’s fat pad in relation to retrocalcaneal problems and the Achilles tendon. J Anat 2006.
- Gibbon WW, Cooper JR, Radcliffe GS. Sonographic incidence of tendon microtears in athletes with chronic Achilles tendinosis. Br J Sports Med 1999.

5. Charcot Neuroarthropathy — Eichenholtz Staging

Charcot neuroarthropathy is a progressive, destructive process affecting the foot and ankle in patients with peripheral neuropathy, most commonly from diabetes. The Eichenholtz classification stages disease through three phases of active bony destruction and repair, and correctly identifying the stage has direct treatment consequences — particularly distinguishing active Charcot from osteomyelitis, since the two can mimic each other closely on imaging and clinically.

Eichenholtz Classification — Charcot Neuroarthropathy

Stage	Findings
0 (Prodromal, per Sanders/Frykberg modification)	Clinical warmth, swelling, and erythema; radiograph may still be normal
I (Fragmentation/Development)	Active bony destruction — cortical disruption, fracture, joint subluxation, debris formation
II (Coalescence)	Decrease in debris; early bone repair and fusion
III (Reconstruction/Consolidation)	Remodeling complete; residual deformity present but the foot is mechanically stable

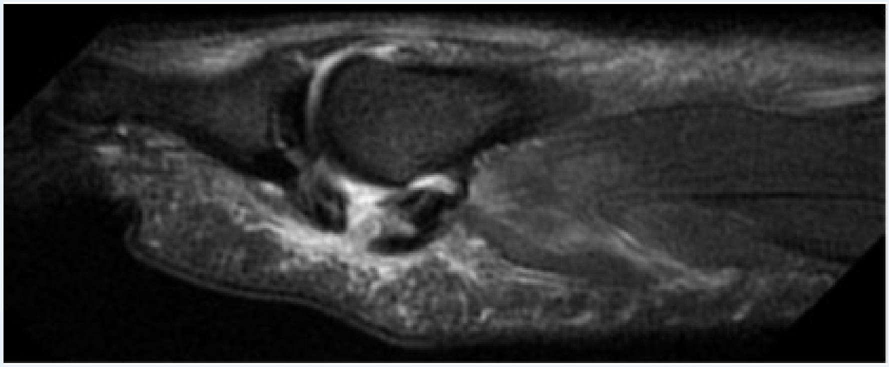
! Active Stage I/II Charcot can closely mimic osteomyelitis both clinically and on imaging — in the right clinical context (diabetic neuropathy, absence of a sinus tract or adjacent ulcer directly overlying the area in question), favor Charcot, but flag the distinction explicitly in the report when genuinely uncertain, since management differs substantially.

6. Turf Toe & Freiberg Infraction

Turf toe refers to injury of the first MTP joint plantar plate and surrounding capsuloligamentous structures from hyperextension, typically on artificial turf. Freiberg infraction is osteonecrosis of a lesser metatarsal head, most often the second, typically affecting adolescents and young adults. Both conditions have established grading systems that guide return-to-play and surgical decision-making respectively.

Turf Toe Grading (First MTP Plantar Plate Injury)

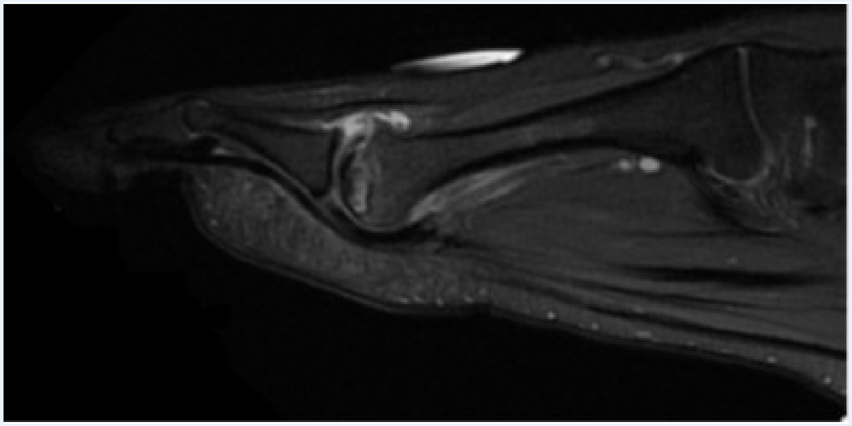
Grade	Finding
I	Plantar plate/capsule sprain; attenuation without discontinuity; no instability
II	Partial tear of the plantar plate/capsuloligamentous complex; localized tenderness; mild instability
III	Complete tear; gross instability; may require surgical assessment and a longer recovery course



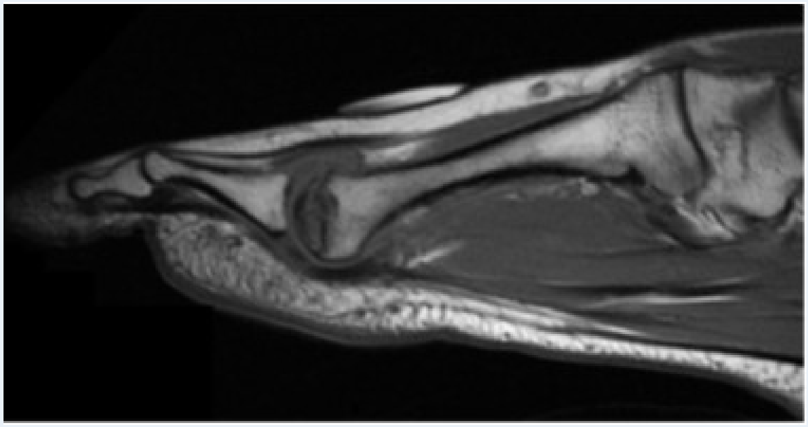
Sagittal MRI of the first MTP joint demonstrating plantar plate disruption in turf toe injury.

Smillie Staging — Freiberg Infraction

Stage	Finding
I	Normal radiograph; subchondral edema on MRI only
II	Widening and flattening of the metatarsal head beginning; no fragmentation
III	Central depression/collapse of the metatarsal head with loose body formation
IV–V	Progressive disintegration of the metatarsal head with secondary MTP arthrosis



Sagittal MRI of Freiberg infraction showing metatarsal head flattening and irregularity.



Corresponding sagittal sequence demonstrating collapse of the metatarsal head articular surface in the same case.

Key References

• Smillie IS. Freiberg’s infraction (Kohler’s second disease). J Bone Joint Surg Br 1957.